

AI for Economic Research: Dynamic Models, Language, and Agents
Lecture 10.8: Case Study 6
AI as Co-Developer:
Ramsey Taxation, Public Debt, and Self-Insurance

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Topic 10.8 Agenda

Research question. Under incomplete markets and idiosyncratic employment risk, what does **Ramsey optimal taxation** imply for **public debt**, **household self-insurance**, and **long-run fiscal dynamics**?

We pursue this by extending a representative-agent deep Ramsey solver (debt sustainability, endogenous admissible sets) to a heterogeneous-agent economy—with AI compressing the model–algorithm–code loop along the way.

1. **The workflow** — homotopy logic applied to this research project
2. **The seed** — representative-agent Ramsey taxation and debt sustainability
3. **Opening the conversation** — how the AI collaboration was set up
4. **Milestones** — refined RA $\rightarrow$ stripped-down HA $\rightarrow$ full HA $\rightarrow$ new discoveries
5. **The final product** — the HA Ramsey model, algorithm, and codebase
6. **Major findings** — debt limits and ergodic dynamics (RA) vs. self-insurance and bistability (HA)
7. **Lessons** — what this paradigm shift demands from the economist

Capstone instance of the Lec09 T5a homotopy workflow—a live project that produced new economics.

Where This Case Study Sits

Lec09 T5a taught the workflow in the abstract. This topic shows it producing new economics.

Layer	What we started with	What we ended with
Economic model	RA Ramsey taxation, incomplete markets (Feng–Han–Zhu)	HA Ramsey with employment risk and borrowing constraints (Chien–Feng–Wen)
Algorithm	Two-level fixed point; admissible set as DNN classifier	Viability kernel via α -shapes; FB complementarity; Lyapunov-type penalty
Code	Working PyTorch RA solver	Full HA research codebase (v1 $\rightarrow$ v2_fullmodel)

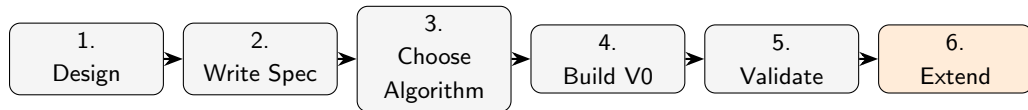
All three layers — model, algorithm, code — were grown *together*, with AI compressing the loop between them. The audit trail lives in `Case_Study/` next to this lecture folder.

1. The Workflow

Homotopy: deform a validated seed toward the research target

Recap: The Homotopy Workflow (Lec09 T5a)

Solve an easier version first, validate it, then continuously deform it toward the full research task.

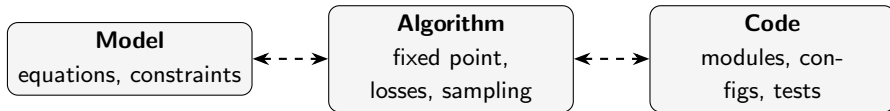


- Start from a version with clear intuition and known checks
- Add one new margin at a time; attribute new bugs to the new margin
- Never ask the agent to “solve the full complicated problem” in one jump

Today’s twist: the “V0” is not a toy — it is a complete, published-quality RA solver. The homotopy runs from one research paper to the next.

Three Layers Deform Together

In computational research, the object being deformed is a triple:



- A new model element (borrowing constraints) forces a new algorithmic element (complementarity penalties), which forces new code (FB residuals in the loss)
- Discoveries also flow *backwards*: a numerical pathology (divergent rollouts) revealed a missing theoretical object (the viability kernel)
- AI's role: keep the three layers *synchronized* — TeX, pseudocode, and PyTorch must always describe the same economy

The homotopy path of this case: RA model/algorithm/code $\rightarrow$ stripped-down HA $\rightarrow$ full HA $\rightarrow$ HA + geometric viability + Lyapunov selection.

What Makes a Good Seed?

The single most important input to this workflow is the starting artifact.

- **It works.** The RA solver trains, simulates, and reproduces the paper's quantitative results — every later failure is attributable to a change
- **It is documented.** A TeX file states the model and the algorithm in the same notation the code uses
- **It is modular.** Sampling, scoring, training, and simulation live in separate modules with a config file — the agent can modify one without breaking the rest
- **It points somewhere.** The RA model is the natural degenerate case of the HA model (shut down idiosyncratic risk and heterogeneity)

A good seed is to agentic research what a good instrument is to empirical work: most of the credibility is purchased before the main analysis starts.

2. The Seed

The representative-agent deep Ramsey solver (Feng–Han–Zhu)

The RA Model in One Slide

Ramsey optimal taxation with non-contingent debt (Feng–Han–Zhu, *Ramsey Taxation, Public Debt Sustainability, and Asset Pricing in Incomplete Markets*).

Representative household, linear technology $Y = L$, flat labor tax τ , one-period bond at price q , Markov spending shock $G(z)$. Recursive Ramsey state: (z, B, λ) with co-state $\lambda = u'(c)$.

$$V(z, B, \lambda) = \max_{\{\lambda_+(z_+)\}} \left\{ u(c) - h(l) + \beta \mathbb{E}[V(z_+, B_+, \lambda_+)] \right\}$$

subject to

$$G(z) + B \leq \tau l + qB_+, \quad c + G(z) = l, \quad (1 - \tau)u'(c) = h'(l), \quad q u'(c) = \beta \mathbb{E}[\lambda_+ | z],$$

and the *sustainability* requirement $(B, \lambda) \in \Omega(z)$.

- $\Omega(z)$: the set of debt/co-state pairs from which the government can stay solvent forever — **endogenous and unknown a priori**
- **Quasi-linear case** $u(c) = c$: $\lambda \equiv 1$, $q = \beta$, state collapses to (z, B) ; the marginal value of debt V_B is a *supermartingale*

The RA Algorithm: Two-Level Fixed Point

Three neural networks trained jointly (PyTorch, actor-critic style):

Network	Object	Training signal
f_Λ	policy: contingent co-states $\lambda_+(z_+)$	maximize simulated welfare + terminal value, penalty $\kappa [1 - f_\Omega(S)]$ off the admissible set
f_V	value function	MSE against realized discounted utility on roll-outs
f_Ω	admissible set Ω	binary classification: label a state 1 if its successor stays in $\Omega^{(n)}$ with finite value

Outer iteration n :

1. Given $\Omega^{(n)}$: update policy and value (rejection-sample states from f_Ω , simulate, fit)
2. Given the new policy: relabel sampled states, retrain $f_\Omega^{(n+1)}$

The key idea to carry forward: domain-finding is itself a fixed-point problem, solved *jointly* with the policy.

The RA Codebase

Five modules plus one config — the structure the agent will inherit:

File	Role
<code>config.json</code>	all economics and hyperparameters: shocks, bounds, penalties, sampling, admissibility thresholds, simulation options
<code>dashboard.py</code>	orchestration: load config, create networks, sample, train
<code>adaptive_sampling.py</code>	admissibility scoring, adaptive sampling, dynamic boundary updates
<code>value_module.py</code>	policy/value networks, trainer, plotting utilities
<code>simulation.py</code>	post-training rollouts and figures
<code>pretrain.py</code>	optional pretraining from legacy solution data

- Configuration-driven design: every later experiment is a config diff, not a code fork
- In the RA case the feasible region is low-dimensional — representable by debt bounds $B_{\min}(\mu, g)$, $B_{\max}(\mu, g)$ estimated within bins
- **This cheap representation is exactly what will break under heterogeneity**

Why This Seed Could Grow

Extension potential, layer by layer:

- *Model*: the RA economy is the no-risk limit of an economy with employed/unemployed households — the HA model nests it
- *Algorithm*: the two-level fixed point (policy $\leftrightarrow$ domain) carries over unchanged in spirit; only the *representation* of Ω must change
- *Code*: actor–critic loop, config plumbing, simulation and plotting all reusable

What is genuinely missing:

- High-dimensional state (5D vs. 2–3D)
- Occasionally binding borrowing constraints (complementarity, not box penalties)
- A geometric representation of the admissible set
- Equilibrium-selection machinery for unit-root dynamics

The gap list above *is* the research plan. Each missing item became a milestone in the AI collaboration — and two of them became new discoveries.

3. Opening the Conversation

Setting up the AI collaboration (from the case-study record)

The Package Is the Context

The collaboration record lives in `Case_Study/` — organized so an agent (or a student) can reconstruct every step:

Folder / file	Role
<code>.../RA_Ramsey_NN_original/</code>	the seed: working RA solver
<code>.../RA_local_refined_v2/</code>	Milestone 1: refined RA code
<code>.../HA_Ramsey_HA_NN/v1/</code>	Milestone 3: first HA prototype
<code>.../HA_Ramsey_HA_NN/v2_fullmodel/</code>	Milestone 4: full HA implementation
<code>02_tex_model_algorithm_growth/</code>	TeX tracking model/algorithm versions
<code>03_notes_and_audits/</code>	code-vs-document audits, dialogue notes
<code>deep_ramsey_ai_case_study.md</code>	the narrative this topic is based on

- Theory TeX and code are kept *side by side* so every equation can be checked in both places
- Conversation provenance (chat links) is recorded, but local files are the citable artifacts

The First Prompts: Restate, Then Audit

The conversation did not start with “build me an HA model.” It started with comprehension and audit tasks on the seed:

1. "Here are the RA paper (TeX) and the RA repository.
Restate the model and the algorithm in your own words.
List every equation and where it is implemented."
2. "Audit the code against the document. Report every place
where the code and the TeX disagree: scoring, bounds,
sampling, penalties. Classify each as a correctness risk
or a documentation gap."
3. "Propose a refined RA document + code pair in which the
two are consistent. Do not add any new economics yet."

- Output of step 2 is preserved: 03_notes_and_audits/code_vs_document_analysis.md
- Mirrors the Lec09 best practice: *ask the agent to restate the spec before it writes code*

Why audit first? Extending an inconsistent seed propagates the inconsistency into every later version.
The audit is the cheapest insurance in the whole project.

The Division-of-Labor Contract

Set expectations explicitly at the start — who owns what:

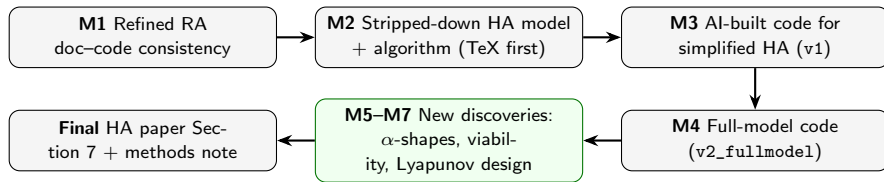
Contribution	AI	Economist
Drafting and reorganizing LaTeX formulations	✓	
Translating equations into vectorized PyTorch	✓	
Proposing FB-penalty and boundary-learning patterns	✓	
Summarizing diagnostics and code/document mismatches	✓	
Checking economic correctness of formulations		✓
Deciding which simplifications are acceptable		✓
Catching conceptual bugs (timing, transition direction)		✓
Setting validation benchmarks; accepting results		✓

The AI supplies **throughput**; the economist supplies **judgment** — the central lesson of the case study, stated before any code was written.

4. Milestones

From refined RA to full HA — and three new discoveries along the way

The Milestone Ladder



- Each milestone is a *validated* artifact preserved in Case_Study/ — the homotopy discipline from Lec09 T5a, applied at research scale
- Green box: where iteration with AI surfaced objects that became **contributions in their own right**

M1: Refine the Seed Before Extending It

First deliverable: a revised RA algorithm and code, with the document and the implementation in exact agreement.

- Artifacts: RA_model_refine_3.tex, deep_ramsey_refined_v3.tex, and the aligned code copy RA_local_refined_v2/
- The audit had found: scoring mismatches, boundary-learning details that differed between TeX and code, and scaling issues
- AI rewrote both sides; the economist adjudicated which side was *right* when they disagreed

Typical fix: the document described admissibility with a smooth power-barrier score; the code used a different functional form. The refined pair uses the same barrier in both — so later HA scores could be derived from a trusted template.

Workflow point: this milestone added *zero* new economics. Resist the urge to skip it.

M2: The New Model, Stripped Down (TeX Before Code)

Add heterogeneity: employed (e) / unemployed (u) households, idiosyncratic employment risk, incomplete markets, borrowing constraints.

Five-dimensional state and three controls:

$$s = (K, a^e, a^u, c^e, c^u), \quad \text{actor outputs } (n^e, c'^e, c'^u)$$

Everything else is derived in one explicit forward pass (separable CRRA, Cobb–Douglas):

$$K' = K^\alpha (\pi^e n^e)^{1-\alpha} + (1-\delta)K - \pi^e c^e - \pi^u c^u, \quad Q = \beta (c^e)^\sigma [(c'^e)^{-\sigma} \pi^{ee} + (c'^u)^{-\sigma} \pi^{ue}]$$

$$a'^e = \frac{1}{Q} \left[\frac{a^e \pi^e \pi^{ee} + a^u \pi^u \pi^{eu}}{\pi^e} + \hat{w} n^e - c^e \right], \quad a'^u = \frac{1}{Q} \left[\frac{a^e \pi^e \pi^{ue} + a^u \pi^u \pi^{uu}}{\pi^u} - c^u \right]$$

Two conceptual bugs the economist caught in AI-drafted, syntactically plausible equations: (1) asset transitions must use *current* consumption c^e, c^u — the budget constraint is a current-period restriction; (2) transition probabilities must follow the *direction of movement* between employment states (π^{ee} vs. π^{ue} etc.). No compiler catches these.

M2 (cont.): Borrowing Constraints as Complementarity

The genuinely new algorithmic ingredient: occasionally binding constraints.

For each type $i \in \{e, u\}$, with Euler discrepancy ϕ^i :

$$a'^i \geq 0, \quad \phi^i \geq 0, \quad \phi^i a'^i = 0$$

Embedded in a differentiable loss via the smoothed **Fischer–Burmeister** function:

$$\Phi_\varepsilon(a, b) = a + b - \sqrt{a^2 + b^2 + \varepsilon^2}, \quad \text{actor loss} = -\mathbb{E}[\text{value}] + \lambda_{FB} \mathbb{E}[\Phi_\varepsilon(\phi^e, a'^e)^2 + \Phi_\varepsilon(\phi^u, a'^u)^2]$$

- λ_{FB} follows an augmented schedule: start moderate (let the policy explore), multiply by ρ whenever average FB residuals stay above tolerance
- The FB residual becomes a *first-class diagnostic*: it measures directly whether complementarity is actually satisfied
- Strategic simplification: log utility, symmetric transitions, simple calibration — **simplicity is a design decision, not a limitation**

M3: AI Builds the Code for the Simplified HA Model (v1)

With the TeX spec fixed, the agent produced the first running HA prototype.

Module	Contents
<code>ha_model.py</code>	actor/critic networks; 5D forward pass; FB residuals; admissibility score
<code>boundary.py</code>	first α -boundary learner (Delaunay + circumradius filter)
<code>dashboard.py</code>	compact loop (≈ 250 lines): boundary update $\rightarrow$ critic $\rightarrow$ actor $\rightarrow$ FB schedule
<code>simulation.py</code> / <code>visualization.py</code>	Monte Carlo rollouts, basic plots
<code>config.json</code>	all parameters, bounds, schedules

- Same module skeleton as the RA seed — deliberate, so diffs stay interpretable
- Validation gate: forward pass reproduces hand-derived transitions; FB residuals shrink under the penalty schedule; admissible region looks economically sensible

Status at M3: a proof of concept, not yet a research instrument.

M4: The Full Model (v2_fullmodel) — Debugging Becomes Architecture

Each numerical failure of v1 was encoded as explicit structure in v2 (dashboard.py grows from ≈ 250 to $> 1,200$ lines):

Failure discovered in v1	v2_fullmodel response
Naive 5D sampling: low admissibility pass rates	adaptive loops keep sampling until enough valid points accumulate
Fragile boundary estimates in 5D	normalized coordinates, Delaunay QJ joggling, save/reload of the α -shape
Rollouts escape the learned admissible set	project_to_admissible (Π_Ω) + projection-distance penalty
Critic bootstrapping instability	frozen target critic with soft updates
“Something is wrong” is not actionable	separate tracking of value, FB, boundary, shape, total losses
Period 0 differs from continuation	explicit period-0 allocation problem in simulation

The explicit **Level-1 (stabilize domain)** / **Level-2 (optimize policy)** loop emerged here — the code finally matched the two-level fixed-point theory of the RA document.

M5 — New Discovery I: α -Shapes for the Feasible Set

Problem: in 5D, debt-bound functions and grids are hopeless. The admissible set must be represented *geometrically* from sampled points.

Solution (Edelsbrunner–Mücke 1994): the α -*shape* of the admissible point cloud — the union of Delaunay simplices with circumradius $\leq 1/\alpha$.

- $\alpha \rightarrow 0$: convex hull; $\alpha \rightarrow \infty$: the bare point set — α tunes how much non-convexity is admitted
- Crucial because the HA feasible set is **generically non-convex** (the borrowing-constraint kink)
- Pipeline: normalize $\rightarrow$ Delaunay $\rightarrow$ circumradius filter $\rightarrow$ find_simplex membership $\rightarrow$ KD-tree buffer

How it surfaced: the agent's first boundary learner kept misclassifying near-kink states. Iterating on *why* led from ad hoc fixes to the computational-geometry literature — a tool import economists rarely make.

Where it lives: boundary.py (AlphaBoundary); written up in the methods note computational_Ramsey.tex and the HA paper, Section 7.

M6 — New Discovery II: Viability Precedes Admissibility

What is the set the α -shape approximates? Iterating with AI on this question connected two literatures.

The admissible set is a **viability kernel** in the sense of Aubin (1991) — the *largest* set of states from which a feasible trajectory exists indefinitely:

$$\Omega = \{s \in H : \exists y \text{ s.t. } s' = f(s, y) \in \Omega \text{ and feasibility/KKT hold}\},$$

$$\mathcal{T}(D) = \{s \in D : \exists y \text{ s.t. } f(s, y) \in D, \text{ constraints hold}\}, \quad D^0 = H, \quad D^{k+1} = \mathcal{T}(D^k) \downarrow \Omega^*.$$

The connection: the same abstract structure as *self-generation* in APS (Abreu–Pearce–Stacchetti, *Econometrica* 1990) — equilibrium sets as the largest fixed point of a monotone set operator, computed by downward iteration; used for Ramsey problems by Phelan–Stacchetti (2001) and Feng (2015). **The math literature on viability precedes the economics literature on admissibility.**

Practical payoff: grid-based set iteration (Saint-Pierre 1994; Feng 2015) is replaced by α -shapes over *unstructured point clouds* — sidestepping the curse of dimensionality.

M7 — New Discovery III: A Lyapunov-Type Penalty for Saddle-Path Selection

Solution: a level-independent Lyapunov-type penalty — flat along the unit-root direction, punishing only movement that signals instability:

$$\mathcal{P}_{\text{lyap}} = \sum_t \beta^t \left[w_{\text{spread}} (c_t^e - c_t^u)^2 + [a_t^{\text{tot}} - a_{t-1}^{\text{tot}}]_+^2 \right]$$

- First term: penalize the *consumption spread*, zero on the $c^e = c^u$ ridge — selects convergence to full insurance without pinning the level
- Second term: penalize *explosive asset growth* (one-sided), leaving the unit-root level free
- Not a formal certificate — a *training-objective* use of Lyapunov logic; to our knowledge new in Ramsey computation

How it surfaced: repeated failed training runs → AI-generated diagnostics isolating the unstable directions → the economist recognized a saddle-path problem → joint design of a penalty with the right invariance properties.

5. The Final Product

The HA model, algorithm, and codebase as they stand

Final Model: HA Ramsey with Employment Risk

Chien–Feng–Wen, Understanding Optimal Ramsey Policies in Heterogeneous-Agents Economies, Section 7.

Environment (quantitative section):

- Employed/unemployed households; ex-ante identical, ex-post heterogeneous
- Symmetric i.i.d. employment risk ($\pi(e|u) = \pi(u|e) = \frac{1}{2}$)
- Borrowing constraint on the unemployed; government bonds the only asset
- Linear technology ($\alpha = \delta = 0$): effective state is 4D, $s = (a^e, a^u, c^e, c^u)$
- Ramsey planner chooses labor tax and debt under full commitment, internalizing households' self-insurance behavior
- **New role for public debt:** not only tax smoothing (RA), but the supply of *liquid assets* for self-insurance

Preferences:

$$u(c, n) = \frac{c^{1-\sigma}}{1-\sigma} - \frac{n^{1+\gamma}}{1+\gamma}$$

CRRA benchmark $\sigma = \gamma = 2$, $\beta = 0.8$; GHH variant for comparison.

Promised consumptions (c^e, c^u) carried as states: the recursive device for Ramsey commitment, inherited from the RA formulation.

Final Algorithm: The Enforcement Stack

Nested fixed point over (actor θ , critic ψ , domain Ω) — the matured version of the RA two-level idea:

Level 1	sample from hypercube H ; score admissibility $\mathcal{A}(s; \theta, \Omega) = w_{geo} \mathbf{1}\{f_\theta(s) \in \Omega\} + w_Q \mathcal{B}(Q^{raw})$; rebuild Ω as the α -shape of high-scoring states
Level 2	form the admissible training pool; critic: TD error on rollouts; actor: welfare — FB complementarity — Euler-sign — projection-distance — Lyapunov-type penalties

- Rollouts are projected back: $s_{t+1} = \Pi_\Omega(f_\theta(s_t))$, with the projection distance penalized
- Amortized updates (domain refreshed every ~ 20 iterations) tame the non-stationarity of learning on a moving domain
- Hyperparameters disciplined by systematic sweeps (32+ configurations) — AI agents ran and summarized the sweeps

Every component answers a specific failure met along the way. The algorithm *is* the accumulated debugging history, made explicit.

Final Codebase and Provenance

Artifact	Status
v2_fullmodel/ code	full HA research code path: versioning, checkpoints, target critic, Level-1/2 loop, period-0 simulation, rich diagnostics
computational_Ramsey.tex	standalone methods note: viability, α -shapes, FB stack, sweep evidence
HA paper, Section 7	quantitative results: transitions, basins of attraction, CRRA vs. GHH
Case_Study/ package	the audit trail: every milestone's code, TeX, and notes

The same triple we started with — model, algorithm, code — but now for a new economy, with genuinely new methodological components and a paper trail connecting every step.

Time from seed to working full HA solver: weeks, not years. The binding constraint was economist judgment at the milestones, not implementation labor.

6. Major Findings

What the seed paper found — and what heterogeneity changes

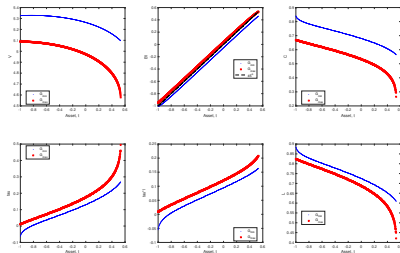
RA Findings I: No Convergence to First Best (Quasi-Linear)

AMSS (2002), quasi-linear + lump-sum transfers:

the government accumulates assets until interest income funds spending — taxes converge to zero (first best); V_B is a martingale converging to 0.

Feng-Han-Zhu: that result is an artifact of lump-sum transfers. Without them, surplus interest income must be rebated through *distortionary* subsidies:

- a “**cost of wealth**” — V becomes non-monotonic in B , V_B turns negative for large assets
- the asset ceiling becomes a **reflecting barrier**, not an absorbing one
- V_B is a *supermartingale*: mean reversion at the ceiling



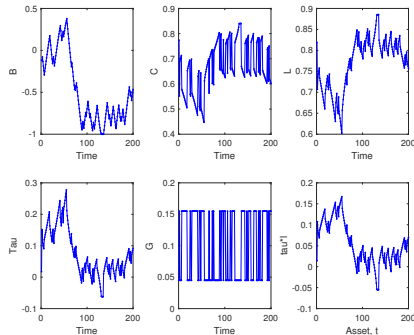
Equilibrium functions ($\underline{B} = -1$): V non-monotonic in debt; V_B changes sign — the cost of wealth. Source: Feng-Han-Zhu.

RA Findings II: Ergodic Debt Dynamics Inside Endogenous Limits

Long-run fiscal behavior in the quasi-linear case:

- Taxes and debt follow **Barro (1979)-style random-walk dynamics** — but confined to endogenous limits $[\underline{B}, \bar{B}]$
- A **unique ergodic stationary distribution** of debt exists (extends Bhandari et al. 2017 to Markov spending)
- The debt ceiling $\bar{B}$ is approached but never reached; the asset ceiling $\underline{B}$ binds with positive probability (mass point)
- No convergence to the complete-markets Lucas–Stokey allocation

Learned jointly with the solution: the admissible set Ω — the ancestor of the viability kernel of Section 4.



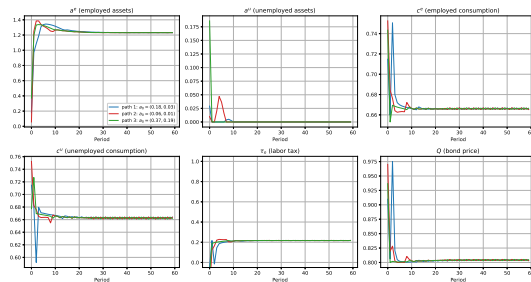
Simulated paths, baseline economy ($\underline{B} = -1$): persistent fluctuations of debt and taxes inside the endogenous limits; the lower bound binds intermittently. Source: Feng–Han–Zhu.

HA Findings I: Front-Loaded Transition to Full Self-Insurance (CRRA)

Section 7, CRRA ($\sigma = \gamma = 2$): the long-run target is **boundary full self-insurance (FSI)** — equal consumption $c^e = c^u = 0.669$, $Q = \beta$, $\tau_n = 0.2$, unemployed constraint binding.

The transition is sharply front-loaded:

- the planner *immediately* issues debt to build the bond stock households need to self-insure
- early borrowing is **cheap**: at $t = 0$ the labor tax is a *subsidy* ($\tau_n < 0$) and bond prices are high
- two reinforcing forces: Lucas–Stokey interest-rate manipulation + the precautionary/liquidity premium ($Q > \beta$); both vanish as FSI is reached



CRRA transition: assets, consumptions, labor tax, and bond price over 60 periods — debt buildup, initial subsidy, convergence to equal consumption with $Q \rightarrow \beta$. Source:

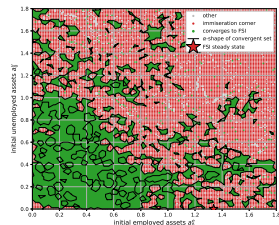
Chien–Feng–Wen, Section 7.

HA Findings II: Bistability — FSI or Immiseration

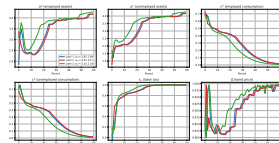
The economy has two long-run outcomes: the FSI steady state and **immiseration** (assets pile up, consumption collapses, $\tau_n \rightarrow 1$).

- Which one is reached depends on **where the economy starts**
- CRRA: only $\approx 13\%$ of initial conditions (a *thin low-asset sliver*) converge to FSI; GHH: $\approx 28\%$
- Basin boundaries are thin and perturbation-sensitive — mapped with the α -shape machinery from Milestone 5

Long-run outcomes are history-dependent: policy alone does not decide between self-insurance and immiseration — initial conditions do.



Basin of attraction in (a_0^e, a_0^u) : green = converges to FSI ($\approx 13\%$).



A divergent (immiseration) path from high initial assets. Source: Chien-Feng-Wen.

What Heterogeneity Adds: RA vs. HA Side by Side

	RA, quasi-linear (seed)	HA, CRRA (new)
Role of public debt	smooth tax distortions over time	<i>also</i> supply liquid assets for household self-insurance
Long-run behavior	ergodic fluctuations inside endogenous debt limits	convergence to full self-insurance — <i>or</i> immiseration
History dependence	initial debt selects a stationary <i>regime</i> (log case)	initial state selects the <i>long-run outcome</i> itself (thin basin)
Interest-rate channel	price effects via q ; negative rates when asset-rich	Lucas–Stokey manipulation of the <i>risk-free</i> rate + precautionary premium
Redistribution	none (representative agent)	transition engineered to lift the unemployed off the borrowing limit
Key constraint object	admissible set $\Omega(z)$ in (B, λ)	viability kernel in 4–5D, non-convex, learned by α -shapes

The new economics is not “RA plus noise.” Heterogeneity changes what debt is *for*, creates a bistable long run, and makes the feasible-set geometry an object of interest.

7. Lessons

A paradigm shift — and what it demands from you

This Is a Paradigm Shift in Economic Research

What changed, concretely, relative to the pre-AI workflow:

- **The theory–code–debug loop compressed** from months to days: drafting, refactoring, auditing, and diagnostics summarization are no longer bottlenecks
- **Documentation became executable**: the TeX spec, the audit notes, and the code evolve in lockstep because the agent keeps them synchronized
- **Cross-literature imports got cheap**: computational geometry (α -shapes), viability theory (Aubin), and control-theoretic stability (Lyapunov) entered an economics project through iteration, not through years of retooling
- **The unit of progress is the validated milestone**, not the finished paper — and every milestone is preserved, reproducible, and auditable

What did *not* change: the AI never decided what was economically interesting, never noticed on its own that an equation had the wrong timing, and never knew when a numerical result was not credible. Those calls remained human at every milestone.

To Benefit the Most, You Need Four Things

1. **A good seed.** A model–algorithm–code triple that already works, is documented, and has the potential to extend toward the problem you want to solve. *(Here: the RA deep Ramsey solver.)*
2. **A clear vision.** Know which problem is interesting, why, and how it builds on what you have. *(Here: what does optimal debt policy look like when households actually need the debt to self-insure?)*
3. **Good judgment.** Steer the research: accept or reject formulations, catch economically wrong but plausible equations, decide which simplifications are strategic and which are fatal. *(Here: the timing bug, the transition-direction bug, the decision to strip down before scaling up.)*
4. **Orchestration.** Run different AI agents/sessions on different layers — formalization, implementation, auditing, sweeps, write-up — and integrate their outputs toward your goal. *(Here: separate conversations for model drafting, code review, debugging, and documentation, all feeding one package.)*

Closing: The Workflow Is the Product

You provide	AI provides
the seed and the destination	speed between them
economic correctness	syntactic and structural correctness
acceptance criteria at each milestone	candidate artifacts to accept or reject
the question “is this credible?”	the diagnostics to answer it

Recap of the path: working RA solver → audit and refine → stripped-down HA model → AI-built prototype → full research code → three new discoveries (α -shapes, viability kernel, Lyapunov design) → new economics (debt as liquidity, bistable long run, history dependence).

Everything is replicable from Case_Study/: the code at every milestone, the TeX at every version, the audits, and the conversation provenance. That package — not any single result — is the template for your own research.

References: Feng–Han–Zhu (Ramsey taxation & debt sustainability); Chien–Feng–Wen (HA Ramsey, Sec. 7); Chien–Feng (computational methods note); AMSS (2002); Barro (1979); Lucas–Stokey (1983); Abreu–Pearce–Stacchetti (1990); Aubin (1991); Saint-Pierre (1994); Edelsbrunner–Mücke (1994); Phelan–Stacchetti (2001); Feng (2015).